



TavaOne Education

General License Course · 2023–2027 Pool

Module 5

Antennas, Feed Lines & Safety

General License Course · G9 + G0 · 6 Exam Questions



Hour 1

HF Antennas and Feed Lines



G9A — SWR and Feed Lines

SWR (Standing Wave Ratio): measure of impedance mismatch between the feed line and the antenna

1:1 SWR = perfect match; all power gets to the antenna. This is the ideal.

At 2:1 SWR: ~11% of power is reflected back; about 89% reaches the antenna — acceptable

At 3:1 SWR: ~25% reflected; most transceivers fold back power above 3:1 to protect their finals

High SWR causes: impedance mismatch, damaged/wet connector, antenna not resonant, wrong feed point

SWR at the transmitter $\neq$ SWR at the antenna: feed line loss absorbs some of the reflected power → makes SWR look better at the radio end (but you're still losing power to heat)

Dummy load (50 ohms): produces 1:1 SWR — use for testing without transmitting into the air

Antenna tuner (ATU): matches the radio output to the feed line; SWR at the radio becomes 1:1; SWR on the feed line between tuner and antenna is unchanged



G9A — Feed Line Comparison

RG-58: small diameter coax; high loss (~ 3 dB/100 ft at 30 MHz); fine for short portable runs, not base station HF

RG-213: standard 50-ohm coax; moderate loss (~ 1.5 dB/100 ft at 30 MHz); good for most base station use

LMR-400: low-loss coax; ~ 0.7 dB/100 ft at 30 MHz; use for long runs or high frequencies

LMR-600/LMR-1200: even lower loss; used for VHF/UHF long runs and contest stations

450-ohm open-wire (ladder line): very low loss (< 0.1 dB/100 ft); balanced; needs balanced tuner input; can work with high SWR without excessive loss

Velocity factor: how fast the signal travels relative to speed of light in free space

RG-58 VF ≈ 0.66 ; RG-213 VF ≈ 0.66 ; LMR-400 VF ≈ 0.85 ; 450-ohm ladder line VF ≈ 0.91

Electrical length = physical length $\times$ velocity factor; matters for phased arrays and matching stubs



G9A — Quarter-Wave Matching Section

Quarter-wave transformer: a section of transmission line exactly $\frac{1}{4}$ wavelength long

It transforms impedance: $Z_{\text{match}} = \sqrt{Z_1 \times Z_2}$

Example: to match 50 Ω coax to a 200 Ω antenna — use a $\frac{1}{4}$ -wave section of 100 Ω line

Because: $Z = \sqrt{50 \times 200} = \sqrt{10000} = 100\Omega$

Physical length of $\frac{1}{4}$ -wave section = $(246/f \text{ in MHz}) \times \text{velocity factor}$

For a 14 MHz signal in 50 Ω coax (VF=0.66): $246/14 \times 0.66 \approx 11.6$ feet

Half-wave section: transforms impedance exactly the same as if not there; useful for putting the match point at a convenient location

End-fed half-wave (EFHW) transformer: 49:1 or 64:1 unun; transforms $\sim 2450\Omega$ antenna impedance to $\sim 50\Omega$; works on multiple bands



G9B — HF Antenna Fundamentals

Half-wave dipole: simplest antenna; resonant; fed at center; length = $468/f(\text{MHz})$ feet total

Example: 20m dipole at 14.225 MHz = $468/14.225 = 32.9$ ft total (16.45 ft per side)

Height matters: higher dipole → lower takeoff angle → better DX; lower → higher angle → regional

Flat-top dipole feeds at $\sim 70\Omega$ (varies with height); matched reasonably well with 50 Ω coax + choke balun

Inverted-V: apex up, ends down at 45°; feedpoint impedance lowers ($\sim 50\Omega$ often); omnidirectional horizontally

EFHW (End-Fed Half-Wave): very popular for portable; 49:1 unun at feedpoint; single conductor runs to tree

EFHW works on fundamental + harmonics: 40m half-wave also resonates on 20m and 10m (even harmonics)

Fan dipole: multiple dipoles fed at the same point; each resonates on its own band; no traps needed



G9C — Directional Antennas

Yagi Antenna

Driven element: fed directly from the coax; $\frac{1}{2}$ -wave dipole

Directors: shorter than driven element; placed in front (toward desired direction); one or more

Reflector: longer than driven element; placed behind; only one

More elements = more gain and narrower beamwidth

Gain in dBd (over dipole): 3-el Yagi $\approx$ 5 dBd; 5-el $\approx$ 8 dBd

$\text{dBi} = \text{dBd} + 2.15$ (isotropic reference = 2.15 dB below dipole)

Baluns for Antennas

1:1 choke balun: blocks common-mode current on coax shield; used at dipole and Yagi feedpoints

Without a choke balun: RF flows on coax shield $\rightarrow$ comes into shack $\rightarrow$ causes RFI

4:1 voltage balun: transforms 200Ω balanced (folded dipole) to 50Ω

4:1 current balun: used with ladder line to coax transition at ATU output

Unun: unbalanced-to-unbalanced; 49:1 for EFHW; 9:1 for end-fed random wire



G9D — Antenna Tuners

Antenna tuner (ATU, transmatch): matches the transceiver output impedance to the feed line+antenna system

What it does: presents a 50Ω match to the transceiver so the radio sends maximum power

What it does NOT do: reduce SWR on the feed line between the tuner and antenna

What it does NOT do: make a bad antenna better; a lossy antenna or lossy feed line stays lossy

What it does NOT do: protect the antenna — it protects the transceiver final

L-network: simple 2-element network; efficient; works best in specific impedance ranges

T-network: most common in commercial tuners (MFJ, LDG, etc.); wide impedance range

Pi-network: used in tube amplifiers; provides some harmonic filtering

Mobile: loading coils on short HF verticals compensate for the antenna being electrically short; efficient is ~50% at best



Hour 2

RF and Electrical Safety



GOA — RF Safety and MPE

RF radiation: non-ionizing; does not break chemical bonds (unlike X-rays); can cause heating at high intensities

MPE (Maximum Permissible Exposure): FCC-established limits for how much RF field strength humans may be exposed to

Two environments: Controlled (operator and people who are aware of RF, like family) vs. Uncontrolled (neighbors, passersby) — different MPE limits

Duty cycle: fraction of time transmitting during a given period; matters because RF exposure is time-averaged

FM: 100% duty cycle (carrier is always present when transmitting)

CW: ~40% duty cycle (key down/up, not continuous)

SSB: ~20% duty cycle (carrier only present during voice peaks, approximately)

FT8 and digital modes: ~100% while transmitting (continuous carrier during each 15-second burst)



G0A — RF Exposure Evaluation

When required: any station that exceeds power levels in Part 97.13(c)(1) must perform an RF exposure evaluation

The thresholds vary by band: e.g., HF below 30 MHz: mobile stations need evaluation if >50W PEP

Evaluation methods: use FCC OET Bulletin 65 tables, online calculators (ARRL RF Safety Calculator), or math

Variables: transmitter power, antenna gain, feed line loss, operating frequency, duty cycle, distance

Reducing exposure: increase distance (power density drops as $1/r^2$), reduce power, reduce duty cycle, use directional antenna (point away from people)

Near-field vs. far-field: within about 1 wavelength = near field; calculations differ; near field can be complex

Practical: most 100W HF stations with base station antennas at reasonable height are fine; document the evaluation

The FCC requires the evaluation be done — NOT that you pass some test; you evaluate and document



G0B — Electrical Safety

Power supply filter capacitors: can hold lethal charge after power is removed; discharge before touching

Bleeder resistor: connected across filter capacitors; drains charge when power is off; automatic safety device

Grounding: three types in a ham station — equipment chassis ground (safety), RF ground (antenna radials), lightning protection ground

Lightning protection: gas discharge arrestors in coax provide a path to ground for direct strikes

Most important lightning protection: disconnect antennas and feed lines during storms — nothing else works as well

Power wiring: fuse at the battery/power supply, as close to the source as practical; protects the wiring

Antenna installation: never install near power lines; if antenna falls, minimum clearance = 1 wire length

Tower climbing: never climb alone; use proper harness; know emergency rescue procedures



Checkpoint Quiz

What is the impedance of a quarter-wave matching section needed to match 50 ohms to 200 ohms?

A. 100 ohms

B. 150 ohms

C. 50 ohms



Checkpoint Quiz

Which of the following actions does an antenna tuner (transmatch) perform?

A. It matches the transmitter output impedance to the feed line and antenna system

B. It reduces SWR on the feed line between the tuner and antenna

C. It increases antenna efficiency by canceling reactive components



Checkpoint Quiz

In a Yagi antenna, where are the director elements located relative to the driven element?

A. In front of the driven element, toward the desired direction of transmission

B. Behind the driven element, away from the desired direction

C. On either side of the driven element for horizontal gain



Checkpoint Quiz

What is the approximate duty cycle of single-sideband voice transmissions?

A. 20%

B. 100%

C. 50%



Checkpoint Quiz

What is the purpose of a bleeder resistor in a power supply?

A. To discharge filter capacitors when power is removed

B. To regulate the output voltage

C. To limit current to protect the rectifier diodes



HamStudy Assignment

Drill these sections before capstone:

G9A — Feed Lines & SWR

G9B — HF Antennas

G9C — Directional Antennas

G9D — Tuners & Special Antennas

G0A — RF Safety & MPE

G0B — Electrical Safety

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